Universal gravitational energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use the zero-at-infinity convention for gravitational potential energy.

SUCCESS CRITERIA - I can...

1. Use center-to-center distance.
2. Explain why negative potential energy does not mean negative kinetic energy.
3. Compute changes from final minus initial energy.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A cart's work is measured over the first **3 m**, but its speed change is measured over **5 m**. Can these values directly test $W_{net}=\Delta K$? Explain.

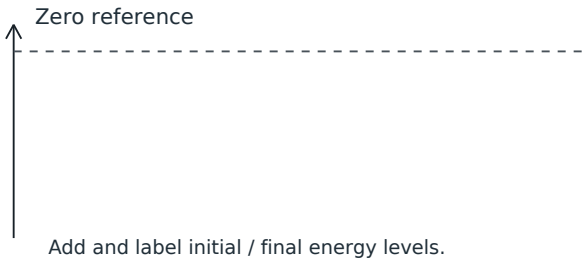
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $U = -GMm/r$; $\Delta U = U_f - U_i$.
Outside a spherical source; **choose U = 0** at infinity. Moving outward makes U less negative.

Worked reasoning: For a supplied source with GMm=**120 J·m**, potential energy is U=**-120**/r. At r=**3 m**, U=**-40 J**; at **6 m**, U=**-20 J**. Moving outward raises U by **20 J**.

Your turn: Draw an energy number line and **mark the direction toward the** zero at infinite separation.



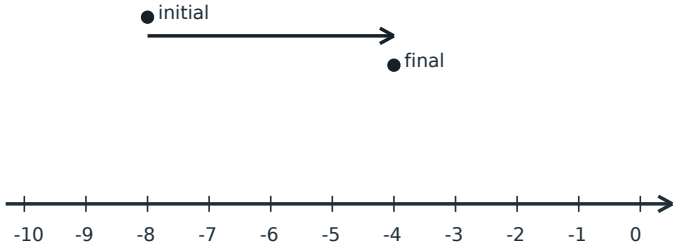
Add and label initial / final energy levels.

Label energies relative to U = 0 at infinite separation.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

An object moves outward from r to **2** r. Does gravitational potential energy become more or less negative? **Draw a number-line comparison**.

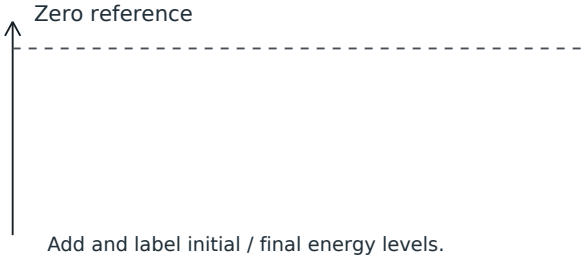


Example: potential energy (J)

Marzano check - circle: ① ② ③

04 Apply the physics

A gravity simulation uses GMm = **24 J·m** and U = **0** at infinite separation. **Find U** at center distances **3 m** and **6 m**, then ΔU for the outward move. Place both energies on the energy-level diagram.



Add and label initial / final energy levels.

Label energies relative to U = 0 at infinite separation.

Marzano check - circle: ① ② ③

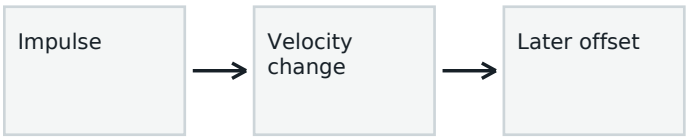
05 Explain your reasoning

A student says “negative potential energy means the object has negative kinetic energy.” Correct the statement.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

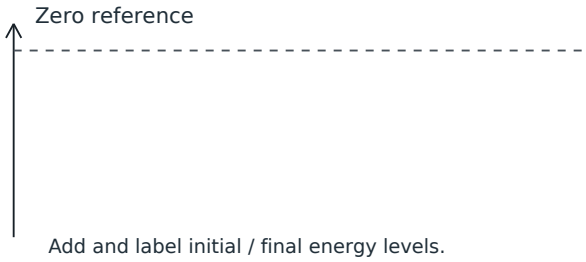
In the same calculation for straight-line deflection, keep sideways Δv fixed but make lead time **4** times as long. **Compare shift** and **name a limitation**.
Label the quantities in the diagram and say what it treats as true.



Marzano check - circle: 1 2 3

07 Exit check

U changes from **-10 J** to **-6 J**. Calculate ΔU and state whether the object gained gravitational energy.

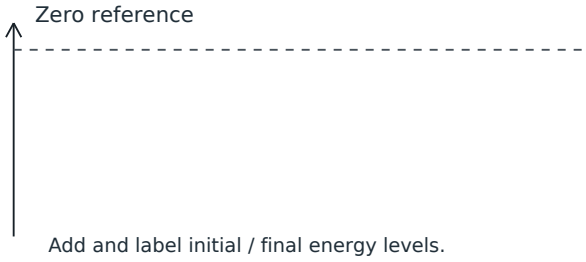


Label energies relative to U = 0 at infinite separation.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Return to the worked example in problem **2**.
1. Use the same numerator to **find U** at **4** and **12 m**.
2. Is the energy change positive or negative when moving outward?



Label energies relative to U = 0 at infinite separation.

Marzano check - circle: 1 2 3